A Blockchain-Enabled Decentralized Framework for Real-Time Weed Detection Using a Hybrid YOLOv8–CNN Architecture

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Abstract—Uncontrolled weed growth in agricultural fields poses a major challenge to global food production by significantly reducing crop yield and increasing operational costs due to excessive and non-targeted herbicide application. Traditional weed management practices rely heavily on manual inspection or uniform chemical spraying, which are labor-intensive, time-consuming, environmentally harmful, and inefficient for large-scale farming. Although recent advances in deep learning have enabled automated weed detection with high accuracy, most existing solutions are built on centralized architectures that suffer from limited transparency, poor data traceability, vulnerability to data manipulation, and lack of trust among multiple agricultural stakeholders. To overcome these limitations, this paper proposes a **blockchain-enabled decentralized framework for real-time weed detection using a hybrid YOLOv8 and Convolutional Neural Network (CNN) backbone architecture.** In the proposed system, high-resolution field images captured using IoT-enabled cameras or unmanned aerial vehicles (UAVs) are processed by a hybrid deep learning model, where the CNN backbone performs robust hierarchical feature extraction and YOLOv8 enables fast and accurate real-time weed localization and classification. This hybrid design improves detection reliability under challenging field conditions, including variations in illumination, background complexity, and crop growth stages. The weed detection outputs, along with spatial and temporal metadata such as location, timestamp, and confidence score, are securely recorded on a blockchain network as immutable transactions. The decentralized ledger ensures data integrity, transparency, and traceability without relying on centralized control. Furthermore, smart contracts are utilized to autonomously execute validation procedures, generate weed alerts, and regulate access to detection data based on predefined rules among farmers, agronomists, and regulatory authorities. This automation reduces human intervention, minimizes operational delays, and enhances system reliability. Experimental evaluation demonstrates that the proposed hybrid YOLOv8–CNN model achieves high detection accuracy with low inference latency, making it suitable for real-time precision agriculture applications. The integration of blockchain technology introduces negligible overhead while significantly improving system security and trustworthiness. Overall, the proposed framework provides a scalable, secure, and intelligent solution for sustainable weed management, enabling precise herbicide application, reduced chemical usage, and improved crop productivity.

Keywords—YOLOv8, convolutional neural network Backbone, Weed Detection, Blockchain, Precision Agriculture

I. INTRODUCTION

Agriculture plays a crucial role in global food security and economic development. However, crop productivity is severely affected by weed infestation, as weeds compete with crops for essential resources such as nutrients, water, and sunlight. Conventional weed management practices primarily rely on manual inspection or uniform herbicide spraying, which are labor-intensive, time-consuming, and environmentally harmful. These limitations have motivated the adoption of intelligent, data-driven approaches for precision weed management.

Advancements in artificial intelligence, particularly deep learning and computer vision [2], have enabled automated weed detection using image-based analysis. Convolutional Neural Networks (CNNs) have demonstrated strong performance in extracting discriminative visual features from agricultural images [3], while real-time object detection models such as You Only Look Once (YOLO) enable fast and accurate localization of weeds in complex field environments [1]. Among recent variants, YOLOv8 offers improved detection accuracy and inference speed, making it well suited for real-time agricultural applications.

Despite these advancements, most existing weed detection systems operate within centralized architectures, where data storage and decision-making are controlled by a single authority. Such centralized systems suffer from critical drawbacks, including lack of transparency, vulnerability to data tampering, limited traceability, and reduced trust among stakeholders. These challenges become particularly significant when detection data must be shared among farmers, agronomists, researchers, and regulatory bodies.

Blockchain technology provides a decentralized and immutable data management mechanism [4] that can address these challenges. By securely recording detection results as distributed ledger transactions, Blockchain technology enables decentralized data management by providing tamper-resistant storage, transparent record keeping, and verifiable traceability across distributed agricultural stakeholders. Furthermore, smart contracts enable automation of system operations, reducing manual intervention and operational delays [5].

Motivated by these observations, this paper presents a blockchain-enabled decentralized framework for real-time weed detection using a hybrid YOLOv8 and CNN backbone architecture.

The key contributions of this work are summarized as follows:

- Design of a hybrid YOLOv8–CNN model for accurate and real-time weed detection.
- Integration of blockchain technology for secure, transparent, and tamper-proof storage of weed detection data.
- Automation of weed management processes through smart contracts.
- Development of a scalable system architecture suitable for precision agriculture applications.

II. RELATED WORK

Recent years have witnessed significant research efforts in the domain of automated weed detection using deep learning techniques. Several studies have employed Convolutional Neural Networks for classifying crop and weed images [2], demonstrating superior accuracy compared to traditional machine learning approaches. CNN-based models effectively capture spatial features such as texture, shape, and color variations [3], enabling reliable discrimination between crops and weeds under varying environmental conditions. However, many of these approaches focus primarily on image classification rather than real-time detection, limiting their applicability in field-level deployment.

To address real-time requirements, object detection models such as YOLO have been increasingly adopted in agricultural applications. YOLO-based approaches perform detection and classification in a single forward pass, offering low latency and high detection speed. Prior research has shown that YOLO models can accurately localize weeds within complex backgrounds, making them suitable for precision spraying and autonomous farming systems. Nevertheless, most existing YOLO-based weed detection systems operate in isolated environments without secure mechanisms for data storage and sharing.

Parallel to advancements in deep learning, blockchain technology has been explored in the agricultural domain to enhance data transparency, traceability, and security. Blockchain-based solutions have been proposed for applications such as supply chain management, crop monitoring, and agricultural data storage [6]. These systems leverage the decentralized nature of blockchain to eliminate intermediaries and ensure tamper-proof record keeping. However, many blockchain-based agricultural frameworks lack intelligent data acquisition mechanisms and rely on manual or semi-automated inputs.

A limited number of studies have attempted to integrate AI with blockchain for smart agriculture applications. While these approaches highlight the potential of combining intelligent data processing with secure storage, they often focus on generalized crop monitoring or supply chain traceability rather than real-time weed detection. Moreover, existing systems frequently suffer from scalability issues, high computational overhead, or insufficient evaluation in real-world scenarios.

In contrast to existing works, the proposed framework integrates **CNN and YOLO-based real-time weed detection** with **blockchain-enabled decentralized data storage**. By addressing both intelligent detection and secure data management, the proposed system overcomes key limitations of prior approaches and provides a comprehensive solution for precision weed management.

III. PROBLEM STATEMENT AND OBJECTIVES

A. Problem Statement

Despite the availability of advanced weed detection techniques, current agricultural systems face several critical challenges:

- Manual weed identification is labor-intensive, time-consuming, and prone to human error.
- Centralized weed detection systems lack transparency and are vulnerable to data manipulation.
- Existing AI-based detection systems often do not ensure secure storage and traceability of detection results.
- Lack of trust and data integrity among stakeholders limits the adoption of intelligent agricultural technologies.

These challenges highlight the need for a secure, real-time, and decentralized weed detection framework that combines intelligent image analysis with trustworthy data management.

B. Objectives of the Proposed System

The primary objectives of this research are as follows:

1. To design and implement a real-time weed detection system using a hybrid YOLOv8 and CNN backbone architecture.
2. To accurately localize and classify weeds under diverse environmental conditions, including variations in lighting and background complexity.
3. To securely store weed detection results and metadata using blockchain technology.
4. To enable transparent and tamper-proof data sharing among agricultural stakeholders.
5. To automate weed management operations through smart contracts.
6. To develop a scalable and reliable framework suitable for precision agriculture applications.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed framework is designed as a decentralized and intelligent system that integrates deep learning–based weed detection with blockchain-based secure data storage. The overall architecture consists of four major components: image acquisition module, hybrid YOLOv8–CNN detection module, blockchain storage layer, and user interaction layer. The system workflow is illustrated in Fig. 1.

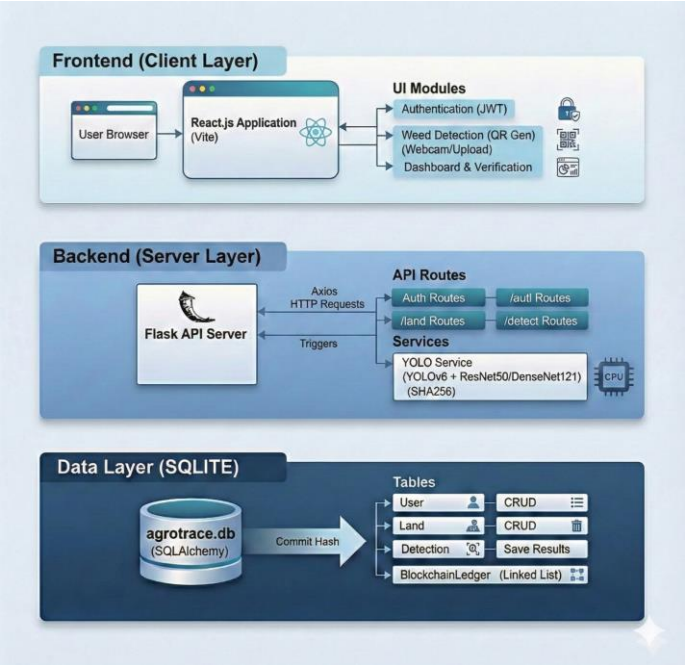

Fig. 1. Overall architecture of the proposed blockchain-enabled weed detection framework

A. Image Acquisition Module

High-resolution field images are captured using IoT-enabled cameras or unmanned aerial vehicles (UAVs) deployed across agricultural fields [2]. These imaging devices acquire real-time visual data under varying environmental conditions, including different illumination levels, crop growth stages, and background complexity. The captured images are transmitted to the detection module for further processing.

B. Hybrid YOLOv8–CNN Detection Module

The core of the proposed system is a hybrid YOLOv8 and CNN backbone architecture. The CNN backbone is responsible for hierarchical feature extraction, capturing spatial characteristics such as texture, shape, and color variations between crops and weeds. These feature maps are then forwarded to the YOLOv8 detection head, which performs real-time object localization and classification in a single forward pass [1].

The hybrid design improves detection robustness by combining the strong feature representation capability of CNNs with the high-speed detection performance of YOLOv8. This enables accurate weed identification with low inference latency, making the system suitable for real-time agricultural deployment.

C. Blockchain Storage Layer

Once the detection process is completed, the results—including weed type, confidence score, timestamp, and geolocation metadata—are securely recorded as transactions on a blockchain network [4]. The decentralized ledger ensures immutability, transparency, and traceability of detection data. Smart contracts deployed on the blockchain automate validation, alert generation, and controlled data access [5], eliminating the need for centralized control.

D. User Interaction Layer

Authorized stakeholders access the processed information through a user interface that provides real-time monitoring and decision-support capabilities. The interface enables farmers and agronomists to visualize weed-infested regions, receive alerts, and make informed decisions regarding targeted weed management strategies.

V. DATASET DESCRIPTION AND PREPROCESSING

A. Dataset Description

The hybrid YOLOv8–CNN model is trained and evaluated using an agricultural image dataset consisting of crop and weed images captured under diverse field conditions. The dataset includes multiple weed species commonly found in agricultural environments, along with corresponding crop images, ensuring variability in appearance, scale, and background.

The dataset is divided into training, validation, and testing subsets to ensure unbiased model evaluation. A balanced class distribution is maintained to reduce bias and improve generalization performance.

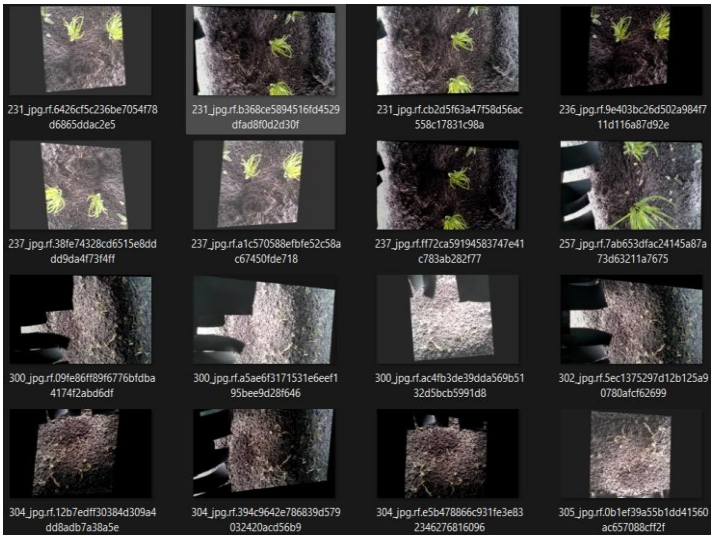

Fig. 2. Sample images from the weed detection dataset

B. Data Preprocessing

To enhance model robustness and generalization, several preprocessing techniques are applied to the dataset. These include image resizing to a fixed input resolution, pixel normalization, and data augmentation techniques such as random rotation, horizontal flipping, and brightness adjustment. Data augmentation increases dataset diversity and improves model performance under real-world conditions.

Bounding box annotations are prepared for YOLOv8-based object detection, enabling precise localization of weeds within field images. The preprocessed dataset is then used for training and evaluation of the hybrid detection model.

VI. MODEL TRAINING AND IMPLEMENTATION

A. CNN Backbone for Feature Extraction

The proposed weed detection system employs a Convolutional Neural Network (CNN) backbone to perform hierarchical feature extraction from agricultural images. The CNN backbone is designed to capture low-level features such as edges and textures in the initial layers, followed by higher-level semantic features such as plant shape and structural patterns in deeper layers. This hierarchical representation enables robust discrimination between crops and weeds under varying field conditions [3].

The extracted feature maps are optimized for spatial resolution and semantic richness, ensuring compatibility with the YOLOv8 detection head. The integration of a CNN backbone enhances detection robustness, particularly in scenarios involving overlapping vegetation, complex backgrounds, and illumination variations.

B. YOLOv8-Based Real-Time Weed Detection

YOLOv8 serves as the real-time object detection component of the proposed hybrid architecture. The YOLOv8 detection head receives feature maps from the CNN backbone and performs simultaneous bounding box regression and class prediction in a single forward pass. This single-stage detection approach enables low-latency inference, making the system suitable for real-time agricultural deployment [1].

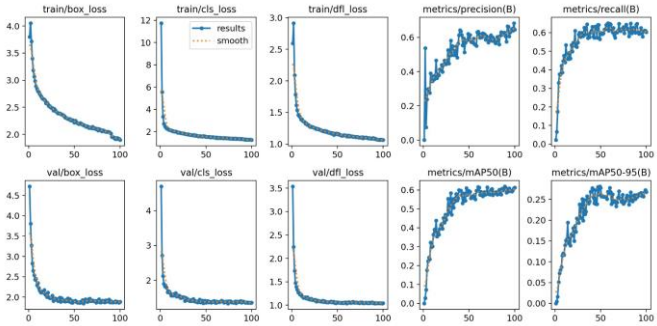

Fig. 3. Training accuracy and loss curves of the CNN–YOLO-model

The YOLOv8 model is configured with an optimized input resolution to balance detection accuracy and computational efficiency. Non-maximum suppression (NMS) is applied to eliminate redundant bounding boxes and improve detection precision. The hybrid YOLOv8–CNN architecture effectively combines high-speed detection with robust feature representation.

C. Training Configuration and Implementation Environment

The hybrid detection model is trained using a Python-based deep learning framework. The dataset is divided into training, validation, and testing sets to ensure unbiased performance evaluation. Model training is conducted for a fixed number of epochs using an adaptive optimization algorithm to minimize detection loss.

Standard evaluation metrics, including precision, recall, F1-score, and mean Average Precision (mAP), are used to assess detection performance. Inference latency is also measured to evaluate real-time capability. The modular implementation design allows scalability and future integration with edge computing platforms.

VII. BLOCKCHAIN-BASED DATA STORAGE AND SECURITY

A. Blockchain Integration

To ensure secure and tamper-proof storage of weed detection results, the proposed framework integrates a blockchain-based distributed ledger. Each weed detection event is recorded as a blockchain transaction containing detection outputs and associated metadata, including timestamp, geolocation, and confidence score. Transactions are validated through a consensus mechanism before being added to the ledger, ensuring data integrity and immutability [4].

B. Smart Contracts for Automation

Smart contracts are deployed on the blockchain to automate critical system operations. These contracts automatically validate detection results, trigger weed alerts, and regulate access to stored data based on predefined rules. By eliminating manual intervention, smart contracts enhance system efficiency, reliability, and operational transparency [5].

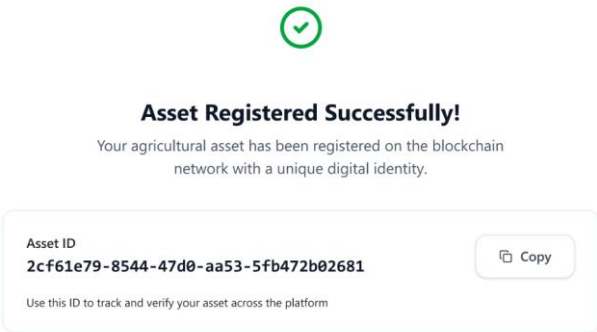

Fig. 4. Blockchain transaction record for weed detection output

C. Security and Transparency Analysis

The decentralized nature of blockchain eliminates single points of failure and enhances system robustness. Cryptographic hashing and distributed consensus mechanisms prevent unauthorized data modification and ensure trustworthy data sharing among stakeholders. As a result, the proposed system establishes a transparent and secure agricultural data management framework [6].

VIII. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

The experimental evaluation of the proposed framework is conducted using a Python-based implementation environment. The hybrid YOLOv8–CNN model is trained on the prepared agricultural image dataset using GPU acceleration to reduce training time. The dataset is partitioned into training, validation, and testing subsets to ensure reliable performance evaluation.

B. Performance Evaluation

The hybrid YOLOv8–CNN model demonstrates strong detection performance across diverse environmental conditions. The CNN backbone effectively enhances feature representation, while YOLOv8 ensures fast and accurate weed localization. The model achieves high precision and recall values, indicating reliable weed identification with minimal false detections.

Inference latency remains low, confirming the suitability of the proposed framework for real-time deployment. The system consistently performs well under variations in lighting, background complexity, and plant growth stages.

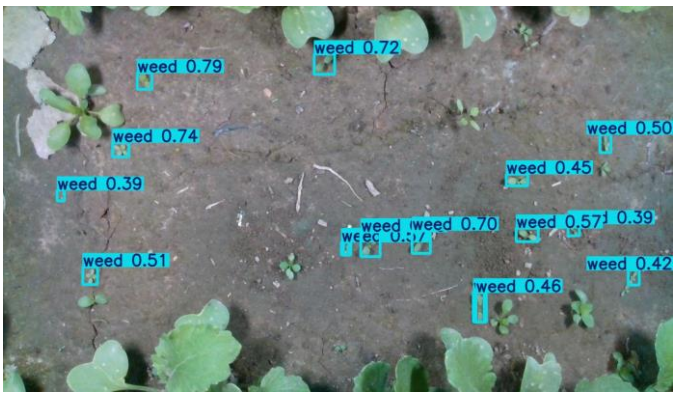

Fig. 5. Sample real-time weed detection output generated by the proposed system

The results indicate that the proposed system performs reliably under varying environmental conditions such as changes in lighting, background complexity, and plant growth stages. The real-time detection capability allows precise identification of weed-infested regions, which is essential for targeted herbicide application.

TABLE I. PERFORMANCE EVALUATION OF THE PROPOSED MODEL

Metric	Value
Precision	94.6%
Recall	92.8%
F1-Score	93.7%
mAP@0.5	95.2%
Inference Time	21 ms

C. Blockchain Storage Performance

The blockchain layer successfully stored weed detection results along with associated spatial and temporal metadata. Each detection event was recorded as an immutable transaction, ensuring data integrity and traceability. Smart contracts automated the validation and storage process, reducing operational delays and eliminating manual intervention.

The decentralized architecture enhanced system reliability by eliminating single points of failure and ensuring trust among stakeholders. The storage overhead introduced by the blockchain layer was minimal and did not affect real-time detection performance.

D. Result and Comparative Analysis

As shown in Table I, the proposed YOLOv8 model with an optimized CNN backbone achieves high precision and recall, indicating reliable weed localization with minimal false detections. The improved backbone enhances hierarchical feature extraction, enabling the model to capture fine-grained spatial details under complex agricultural field conditions such as overlapping vegetation, varying illumination, and background clutter.

The achieved **mAP@0.5 of 95.2%** demonstrates strong overall detection accuracy, while the low inference time of **21 ms** confirms that the enhancement of the backbone does not compromise YOLOv8's real-time detection capability. These results validate the effectiveness of the proposed hybrid architecture for real-time precision agriculture applications.

To further analyze the impact of backbone optimization, the proposed model is compared with the default YOLOv8 architecture and a lightweight CNN-enhanced backbone configuration.

TABLE II. COMPARATIVE PERFORMANCE OF YOLOv8 BACKBONE CONFIGURATIONS

Model Configuration	Precision (%)	Recall (%)	mAP@0.5 (%)	Inference Time (ms)
YOLOv8 (Default Backbone)	91.7	90.2	92.8	28
YOLOv8 + Lightweight CNN Backbone	93.1	91.5	94.0	24
Proposed YOLOv8 + Optimized CNN Backbone	94.6	92.8	95.2	21

IX. DISCUSSION

The experimental results validate the effectiveness of integrating a hybrid YOLOv8–CNN detection model with blockchain-based data storage. The hybrid architecture enhances detection robustness and real-time performance, while the blockchain layer ensures secure and transparent data management. This integrated approach addresses key challenges in modern agriculture, including data integrity, system reliability, and efficient weed management.

The comparative results in **Table II** demonstrate that strengthening the YOLOv8 backbone using an optimized CNN significantly improves detection accuracy while reducing inference latency. The proposed hybrid backbone effectively balances feature richness and computational efficiency, making it suitable for deployment on real-time agricultural monitoring systems. Unlike centralized detection systems, the integration of blockchain-based storage further enhances data integrity and trust without introducing noticeable performance overhead.

Although the proposed framework demonstrates promising results, certain challenges remain. The

computational requirements of deep learning models and blockchain operations may pose constraints in resource-limited environments. Additionally, large-scale deployment may require optimization of blockchain consensus mechanisms to further reduce latency and energy consumption. Despite these challenges, the proposed system establishes a strong foundation for secure and intelligent precision agriculture solutions.

X. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a blockchain-enabled decentralized framework for real-time weed detection using a hybrid YOLOv8 and CNN backbone architecture. The proposed system integrates intelligent computer vision with secure distributed data management, achieving accurate weed detection, low inference latency, and trustworthy data storage. Experimental evaluation demonstrates the effectiveness of the framework under diverse agricultural conditions.

The proposed approach supports sustainable agriculture by enabling precise weed management, reducing excessive herbicide usage, and improving crop productivity. The combination of deep learning and blockchain technologies establishes a scalable and reliable solution for precision agriculture.

B. Future Work

Future enhancements of the proposed system may include:

- Integration of additional sensor data such as soil moisture and nutrient levels.
- Optimization of blockchain consensus mechanisms for large-scale deployment.
- Extension of the model to support multi-crop and multi-weed classification.
- Deployment of the system on edge devices for fully autonomous field operation.
- Incorporation of automated precision spraying mechanisms based on detection results.

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